Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (Bsc.IT Semester III)

Data Structures Practical

Practical - VII

Roll No. : S010	Name: ANJALI GUPTA
Class : SYIT	Batch : 1
Date of Assignment : 05/08/2026	Date/Time of Submission : 05/08/2026

Hash Table:

Write a program to:

Implement a hash table with separate chaining for collision handling.

CODE:

File Edit Format Run Options Window Help

```
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

insert(10,"Apple")
insert(14,"banana")
insert(15,"mango")
insert(20,"grapes")
insert(32,"Watermelon")

print(table)
```

OUTPUT:

```
========================================== RESTART: C:/Users/info/Downloads/Anjalisyit/ds_pra7.py ========
[[(10, 'Apple'), (15, 'mango'), (20, 'grapes')], [], [(32, 'Watermelon')], [], [(14, 'banana')]]
```

b.Store and retrieve data from the hash table.

CODE:

```
File   Edit   Format   Run   Options   Window   Help
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not found"


insert(10,"Apple")
insert(14,"banana")
insert(15,"mango")
insert(20,"grapes")
insert(32,"Watermelon")

print(table)

key = int(input("Enter key: "))
print(search(key))
```

OUTPUT:

```
========================================== RESTART: C:/Users/info/Downloads/Anjalisyit/ds_pra7.py ======
[[(10, 'Apple'), (15, 'mango'), (20, 'grapes')], [], [(32, 'Watermelon')], [], [(14, 'banana')]]
Enter key: 15
mango

========================================== RESTART: C:/Users/info/Downloads/Anjalisyit/ds_pra7.py ======
[[(10, 'Apple'), (15, 'mango'), (20, 'grapes')], [], [(32, 'Watermelon')], [], [(14, 'banana')]]
Enter key: 14
banana
```

Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?